Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

What curve does the locus represent if the locus of a moving point which is always equidistant from the points $(2, -1)$ and $(3, 2)$?

(A) Hyperbola

(B) Straight line

(C) Ellipse

(D) Circle

Q.2

The locus of the point of intersection of the lines $x \sin\theta + (1 - \cos\theta)y = a \sin\theta$ and $x \sin\theta - (1 + \cos\theta)y + a \sin\theta = 0$ is

(A) $x^2 - y^2 = a^2$

(B) $x^2 + y^2 = a^2$

(C) $y^2 = ax$

(D) none of these

Q.3

Let $A(2, -3)$ and $B(-2, 1)$ be vertices of a triangle ABC. If the centroid of this triangle moves on the line $2x + 3y = 1$, then the locus of the vertex C is the line

(A) $3x + 2y = 5$

(B) $2x - 3y = 7$

(C) $2x + 3y = 9$

(D) $3x - 2y = 3$

Q.4

Locus of mid point of the portion between the axes of $x \cos\alpha + y \sin\alpha = p$, where p is constant is

(A) $x^2 + y^2 = 4/p^2$

(B) $x^2 + y^2 = 4p^2$

(C) $1/x^2 + 1/y^2 = 2/p^2$

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(D) $1/x^2 + 1/y^2 = 4/p^2$

Q.5

Locus of the centroid of the triangle whose vertices are $(a\cos t, a\sin t), (b\sin t, -b\cos t)$ and $(1, 0)$; where t is a parameter is

(A) $(3x-1)^2 + (3y)^2 = a^2 + b^2$

(B) $(3x+1)^2 + (3y)^2 = a^2 + b^2$

(C) $(3x+1)^2 + (3y)^2 = a^2 - b^2$

(D) $(3x-1)^2 + (3y)^2 = a^2 - b^2$

Q.6

If $P(1, 0), Q(-1, 0)$ and $R(2, 0)$ are three given points. The point S satisfies the relation $SQ^2 + SR^2 = 2SP^2$. The locus of S meets PQ at the point

(A) $(0, 0)$

(B) $(2/3, 0)$

(C) $(-3/2, 0)$

(D) $(0, -2/3)$

Q.7

If the point $(3, 4)$ lies on the locus of the point of intersection of the lines $x\cos\alpha + y\sin\alpha = a$ and $x\sin\alpha - y\cos\alpha = b(\alpha$ is a variable), the point (a, b) lies on the line $3x - 4y = 0$ then $|a + b|$ is equal to

(A) 1

(B) 7

(C) 12

(D) 5

Q.8

Locus of the mid-points of the intercepts between the coordinate axes by the lines passing through $(a, 0)$ does not intersect

(A) x-axis

(B) y-axis

(C) $y = x$

(D) $y = a$

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Q.9

Locus of mid point of the portion between the axes of $x\cos\alpha + y\sin\alpha = p$ whre p is constant is

(A) $x^2 + y^2 = \frac{4}{p^2}$

(B) $x^2 + y^2 = 4p^2$

(C) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{2}{p^2}$

(D) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$

Q.10

If the sum of the distances of a point from two perpendicular lines in a plane is 1, then its locus is

(A) square

(B) circle

(C) straight line

(D) two intersecting lines

Q.11

Locus of centroid of the triangle whose vertices are $(a\cos t, a\sin t), (b\sin t, -b\cos t)$ and $(1, 0)$, where t is a parameter, is

(A) $(3x+1)^2 + (3y)^2 = a^2 - b^2$

(B) $(3x-1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x-1)^2 + (3y)^2 = a^2 + b^2$

(D) $(3x+1)^2 + (3y)^2 = a^2 + b^2$

Q.12

The line $L_1 : 4x + 3y - 12 = 0$ intersects the x - and y -axis at A and B, respectively. A variable line perpendicular to L_1 intersects the x - and the y axis at P and Q, respectively. Then the locus of the circumcentre of triangle ABQ is

(A) $3x - 4y + 2 = 0$

(B) $4x + 3y + 7 = 0$

(C) $6x - 8y + 7 = 0$

(D) None of these

Q.13

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Vertices of a variable triangle are $(3,4),(5\cos\theta,5\sin\theta)$ and $(5\sin\theta,-5\cos\theta)$, where $\theta\in R$. Locus of it's orthocentre is

(A) $(x+y-1)^2+(x-y-7)^2=100$

(B) $(x+y-7)^2+(x-y-1)^2=100$

(C) $(x+y-7)^2+(x+y-1)^2=100$

(D) $(x+y-7)^2+(x-y+1)^2=100$

Q.14

A variable line ' L ' is drawn through $O(0,0)$ to meet the lines $L_1: y-x-10=0$ and $L_2: y-x-20=0$ at the points A and B respectively. A point P is taken on ' L ' such that $\frac{2}{OP}=\frac{1}{OA}+\frac{1}{OB}$. Locus of $'P$ is

(A) $3x+3y=40$

(B) $3x+3y+40=0$

(C) $3x-3y=40$

(D) $3y-3x=40$

Q.15

A variable line $\frac{x}{a}+\frac{y}{b}=1$ slides on the coordinate axes in second quadrant such that the sum of the length of intercepts made by it on the axes is $\mathbf{k}$, then the locus of the circumcentre of triangle made by the line with the axes is

(A) $2(x-y)+\mathbf{k}=0$

(B) $2(x+y)+\mathbf{k}=0$

(C) $2(x+y)-k=0$

(D) $2(x-y)-k=0$

Answers & Solutions

Q.1 Answer:

Straight line

Solution:

Clarification: Let $\mathrm{A}(2,-1)$ **and** $\mathrm{B}(3,2)$ **be the given points and** (x,y) **be the co-ordinates of a point** P **on the required locus.**

Then, $PA^2=(x-2)^2+(y+1)^2$ **and** $PB^2=(x-3)^2+(y-2)^2$

By problem, $\mathrm{PA}=\mathrm{PB}$ **or,** $\mathrm{PA}^2=\mathrm{PB}^2$

or, $(x-2)^2+(y+1)^2=(x-3)^2+(y-2)^2$

or, $x^2-4x+4+y^2+2y+1=x^2-6x+9+y^2-4y+4$

or, $2x+6y=8$ **or,** $x+3y=4\ldots\ldots\ldots$ **(1)**

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Clearly, equation (1) is a first degree equation in x and y; hence, the locus of P is a straight line whose equation is $x + 3y = 4$.

Q.2 Answer:

$x^2 + y^2 = a^2$

Solution:

From the given equations we have $\frac{1-\cos\theta}{\sin\theta} = \frac{a-x}{y}$ and $\frac{1+\cos\theta}{\sin\theta} = \frac{a+x}{y}$

Multiplying we get $\frac{1-\cos^2\theta}{\sin^2\theta} = \frac{a^2-x^2}{y^2} \Rightarrow x^2 + y^2 = a^2$

Q.3 Answer:

$2x + 3y = 9$

Solution:

Let $C(h, k)$ be the vertex, then the centroid is $\left(\frac{h+2-2}{3}, \frac{k-3+1}{3}\right)$ i.e. $(h/3, (k-2)/3)$ which lies on $2x + 3y = 1$

$\Rightarrow \quad 2\frac{h}{3} + \frac{3(k-2)}{3} = 1$

$\Rightarrow \quad 2h + 3k = 9$ and the locus of (h, k) is $2x + 3y = 9$.

Q.4 Answer:

$1/x^2 + 1/y^2 = 4/p^2$

Solution:

If (h, k) is the mid-point, then $h = p/2\cos\alpha, k = p/2\sin\alpha$

so $\quad (p/2h)^2 + (p/2k)^2 = \cos^2\alpha + \sin^2\alpha = 1$

$\Rightarrow \quad 1/h^2 + 1/k^2 = 4/p^2$

Locus of (h, k) is $1/x^2 + 1/y^2 = 4/p^2$

Q.5 Answer:

$(3x-1)^2 + (3y)^2 = a^2 + b^2$

Solution:

If (h, k) is the centroid, then

$$h = \frac{a\cos t + b\sin t + 1}{3}$$

$$k = \frac{a\sin t - b\cos t + 0}{3}$$

$\Rightarrow (3h-1)^2 + (3k)^2 = (a\cos t + b\sin t)^2 + (a\sin t - b\cos t)^2$

$= a^2 + b^2$

Locus of (h, k) is $(3x-1)^2 + (3y)^2 = a^2 + b^2$

Q.6 Answer:

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$(-3/2, 0)$

Solution:

Let S be the point (x, y) then $(x+1)^2 + y^2 + (x-2)^2 + y^2 = 2\left[(x-1)^2 + y^2\right]$
$\Rightarrow 2x + 3 = 0$, the locus of S and equation of PQ is $y = 0$. So the required points is $(-3/2, 0)$.

Q.7 Answer:

7

Solution:

Squaring and adding the given equations of the lines we get $x^2 + y^2 = a^2 + b^2$ as the locus of the point of intersection of these lines.
Since $(3, 4)$ lies on the locus, we get $9 + 16 = a^2 + b^2$ i.e. $a^2 + b^2 = 25$ (i)
Also, (a, b) lies on $3x - 4y = 0$
so $3a - 4b = 0 \quad \Rightarrow \quad b = (3/4)a$ (ii)
From (i), $a^2 + (9/16)a^2 = 25 \Rightarrow \quad a^2 = 16$
So that $|a + b|^2 = (7/4)^2 a^2 = 49 \quad |a + b| = 7$

Q.8 Answer:

y-axis

Solution:

Equation of any line through $(a, 0)$ be
$\frac{x}{a} + \frac{y}{b} = 1$, where b is a parameter.
This line meets y-axis at $(0, b)$ and if (h, k) denotes the mid-point of the intercept of the line between the coordinate axes,
then $h = a/2, k = b/2$ and then the locus of (h, k) is $x = a/2$.
This clearly does not intersect y-axis.

Q.9 Answer:

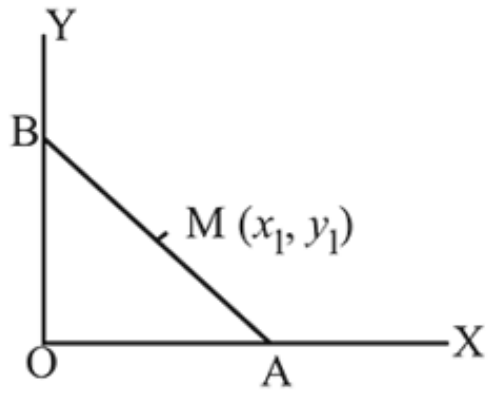
$\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$

Solution:

Equation of AB is $x\cos\alpha + y\sin\alpha = p$
$\Rightarrow \frac{x\cos\alpha}{p} + \frac{y\sin\alpha}{p} = 1;$

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$\Rightarrow \frac{x}{p/\cos\alpha} + \frac{y}{p/\sin\alpha} = 1$



So co-ordinates of A and B are
$\left(\frac{p}{\cos\alpha}, 0\right)$ **and** $\left(0, \frac{p}{\sin\alpha}\right)$
So co-ordinates of midpoint of AB are
$\left(\frac{p}{2\cos\alpha}, \frac{p}{2\sin\alpha}\right) = (x_1, y_1)$ **(say)**
$x_1 = \frac{p}{2\cos\alpha} \& y_1 = \frac{p}{2\sin\alpha}$
$\Rightarrow \cos\alpha = p/2x_1$ **and** $\sin\alpha = p/2y_1$;
Consider $\cos^2\alpha + \sin^2\alpha = 1$
$\Rightarrow \frac{p^2}{4}\left(\frac{1}{x_1^2} + \frac{1}{y_1^2}\right) = 1$
$\therefore$ **Locus of** (x_1, y_1) **is** $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$.

Q.10 Answer:

square

Solution:

Let the two perpendicular lines be the coordinate axes and let the point be $\mathrm{P(h, k)}$**. Then sum of the distances of** $\mathrm{P(h, k)}$ **from the coordinate axes is** $|\mathrm{h}| + |\mathrm{k}|$**. It is given that** $|\mathrm{h}| + |\mathrm{k}| = 1$
So, locus of (h, k) **is** $|\mathrm{x}| + |\mathrm{y}| = 1$**. This gives four lines** $x + y = 1, x - y = 1, -x + y = 1, -x - y = 1$ **which enclose a square.**

Q.11 Answer:

$(3x - 1)^2 + (3y)^2 = a^2 + b^2$

Solution:

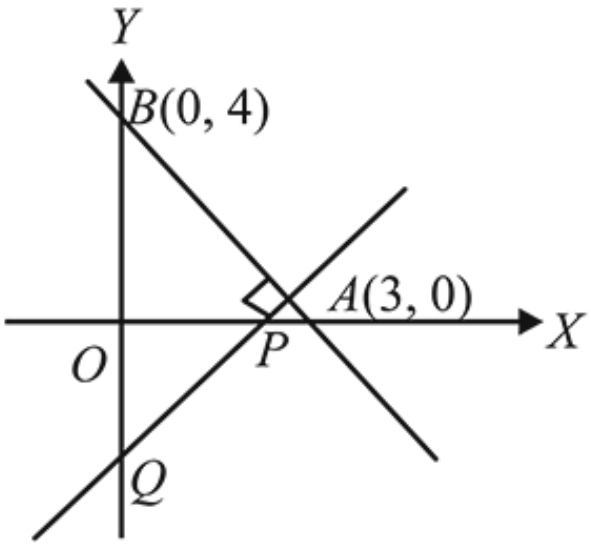
$x = \frac{a\cos t + b\sin t + 1}{3}$
$\Rightarrow a\cos t + b\sin t = 3x - 1$
$y = \frac{a\sin t - b\cos t}{3} \Rightarrow a\sin t - b\cos t = 3y$
Squaring $\&$ **adding,** $(3x - 1)^2 + (3y)^2 = a^2 + b^2$

Q.12 Answer:

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$6x - 8y + 7 = 0$

Solution:



Clearly, the circumcenter of triangle ABQ will lie on the perpendicular bisector of line AB. Now, the equation of perpendicular bisector of line AB is $3x - 4y + 7/2 = 0$. Hence, the locus of circumcenter is $6x - 8y + 7 = 0$.

Q.13 Answer:

$(x + y - 7)^2 + (x - y + 1)^2 = 100$

Solution:

Distance of all the points from $(0, 0)$ are 5 unit. That means circumcentre of the triangle formed by the given point is $(0, 0)$. If $G(h, k)$ be the centroid of triangle, then $3h = 3 + 5(\cos\theta + \sin\theta), 3k = 4 + 5(\sin\theta - \cos\theta)$

If $H(\alpha, \beta)$ be the orthocentre, then $OG : GH = 1 : 2 \Rightarrow \alpha = 3h, \beta = 3k$

$\cos\theta + \sin\theta = \frac{\alpha - 3}{5}, \sin\theta - \cos\theta = \frac{\beta - 4}{5}$

$\Rightarrow \sin\theta = \frac{\alpha + \beta - 7}{10}, \cos\theta = \frac{\alpha - \beta + 1}{10}$

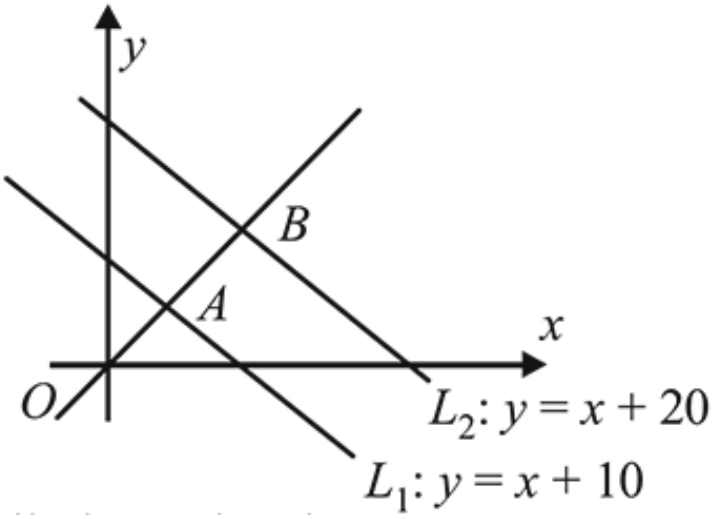
Thus, locus of (α, β) is $(x + y - 7)^2 + (x - y + 1)^2 = 100$.

Q.14 Answer:

$3y - 3x = 40$

Solution:

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Let the parametric equation of drawn line is

$\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r \Rightarrow x = r\cos\theta, y = r\sin\theta$

Putting it in $'L_1'$**, we get** $r\sin\theta = r\cos\theta + 10$

Similarly, putting the general point of drawn line is the equation of L_2**, we get** $\frac{1}{OB} = \frac{\sin\theta - \cos\theta}{20}$

Let $P = (h, k)$ **and** $OP = r$

$\Rightarrow r\cos\theta = h, r\sin\theta = k$**, we have**

$\Rightarrow 40 = 3r\sin\theta - 3r\cos\theta \Rightarrow 3y - 3x = 40$

Q.15 Answer:

$2(x - y) + \mathbf{k} = 0$

Solution:

Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$.

$\because |a| + |b| = \mathbf{k} \Rightarrow -a + b = \mathbf{k}$

$\Rightarrow \mathbf{b} - \mathbf{a} = \mathbf{k}$ **and circumcentre is given by** $\left(\frac{a}{2}, \frac{b}{2}\right)$

$\therefore$ **req. locus is** $2(x - y) + \mathbf{k} = 0$

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